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Phys-EdiGAN: A privacy-preserving method for editing physiological signals in facial videos

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ABSTRACT

Remote Photoplethysmography (rPPG) technology can derive physiological signals such as heart rate by detecting subtle color changes in facial videos, thereby assessing people's physiological and health conditions. However, since this technology can extract physiological data through ordinary cameras without the subject's awareness, it poses a risk of misuse for monitoring and analyzing an individual's physiological and psychological state, potentially leading to significant privacy breaches. To address this issue, this paper introduces Phys-EdiGAN, a facial video physiological signals editing method based on Conditional Generative Adversarial Networks (CGANs). This method enables customizable editing of physiological privacy information in facial videos, effectively preventing privacy leaks. Phys-EdiGAN introduces beneficial noise, allowing the model to learn decoupled representations between rPPG signal features and other features, thereby improving the editing of the rPPG signal in the video. We evaluated Phys-EdiGAN on several popular databases (e.g., UBFC-rPPG and VIPL-HR), and the results demonstrate that it allows precise customization of physiological privacy information while preserving high video quality and ensuring that facial identity, expressions, and movements remain consistent. This technology enhances user trust in facial video monitoring, prevents misuse of physiological data, and improves the overall security of video monitoring systems.

1. Introduction

Facial video technology has been widely applied in various fields such as security surveillance, identity verification, smart homes, and social media. While it has brought many conveniences, it has also raised numerous security and privacy issues, such as personal identity information leakage, unauthorized surveillance, and the misuse of deepfake technology. Additionally, through video analysis technology, it is possible to detect changes in facial blood flow that correspond with the heart cycle, known as remote photoplethysmography (rPPG). This technology can extract physiological signals, including heart rate, pulse, and blood oxygen levels, through cameras without the subject's awareness, thereby compromising their physiological privacy. If this physiological privacy information is obtained by criminals, it could be used for blackmail, fraud, or exploitation, leading to financial losses and psychological distress for individuals, and affecting their normal lives and careers development. Therefore, protecting physiological privacy information is crucial to prevent these negative consequences.

However, current research on protecting physiological privacy in facial videos is very limited. Chen et al. [1] were the first to propose an algorithm capable of eliminating physiological information from facial videos, which is based on motion component magnification. However, their algorithm was only tested on a self-collected dataset, where the only variations were in skin color and facial makeup of the subjects. The generalizability of this algorithm under other interfering factors, such as subject movement and lighting changes, has yet to be verified. Li et al. [2] generated sine wave signals and used the facial region of interest (ROI) to create blurred templates, which were then superimposed onto the source video. However, such extensive pixel-level disturbance may cause issues with visual perception. M.

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Chen et al. [3] developed an algorithm called PulseEdit to remove and modify rPPG signals in videos. The algorithm computes a perturbation signals based on the source rPPG and target signals, applying skin pixel perturbations to the facial region to embed target rPPG signals into the video. Nonetheless, this approach exposes two limitations: (1) PulseEdit introduces a hyperparameter, to balance between editing effectiveness and perceptual distortion. The optimal value of the hyperparameter varies across different lighting conditions, skin tones, and motion scenes, requiring manual tuning; (2) PulseEdit's modifications occur only at the pixel level, making them more susceptible to detection by forensic tools. Furthermore, Z. Sun et al. [4] applied projected gradient descent to alter the facial pixels. However, this method lacks universality, as each video requires separate gradient descent training, resulting in significant time consumption.

Although the aforementioned methods can modify physiological signals in facial videos to some extent, they all share a common characteristic, e.g., directly altering the pixel-level information of physiological signals while leaving other regions of the image unchanged. In contrast, this paper introduces an adaptive physiological signals editing approach based on Conditional Generative Adversarial Networks (CGANs), which has the following advantages.

- Our method can generate more natural and realistic modifications of physiological signals, avoiding artifacts and unnatural visual effects that may arise from direct pixel-level modifications.
- Our method can maintain the overall consistency and quality of the video, reducing local inconsistencies and visual distortions that may result from direct modification methods.
- Our method have strong adaptability, capable of handling variations in different video contexts and being applicable to large-scale datasets and video content.

In our method, we treat the rPPG signals as the label of the video, enabling the network to learn the mapping from the facial video to the corresponding rPPG signals. At the same time, we introduce beneficial noise to positively incentivize the model [5,6], enabling it to decouple the rPPG features from other features of the video. Subsequently, the learned rPPG signals is regarded as a condition, which can be synthesized with any given facial video to generate a new facial video containing the target rPPG signals based on a well-designed conditional GAN model. To ensure that the identity information in the source facial video remains unchanged, we designed an Identity Preservation Component to enforce identity constraints. Due to the powerful capabilities of GANs in generating facial images, our method enables a range of personalized applications, such as rPPG signals erasure, rPPG signals retargeting, and rPPG signals swapping between different faces.

Our main contributions are summarized as follows:

- This paper presents Phys-EdiGAN, a novel method for editing physiological signals in facial videos using Conditional GAN. The method enables precise customization of physiological privacy information, effectively preventing privacy leaks while preserving high video quality and ensuring that facial identity remains consistent.
- We propose a Self-Critic Module to learn the latent mapping between rPPG signals and video frames, guiding the generator to embed target rPPG signals into facial videos. An Identity Preservation Component is added to maintain the person's identity in the generated videos.
- Our experiments on databases such as UBFC-rPPG and VIPL-HR demonstrate that Phys-EdiGAN can generate more natural and realistic modifications of physiological signals, maintain overall video consistency and quality, and is highly adaptable, allowing for a range of personalized applications.

2. Related work

2.1. rPPG technology

As awareness of health issues increases alongside the demand for medical technology, rPPG is gaining traction due to its non-invasive nature and user-friendly application. This technology enables patients to monitor their health without physical constraints, thereby reducing the risk of medical infections, enhancing patient comfort, and facilitating continuous monitoring of physiological parameters, which can aid in the early detection of potential health issues.

Prior to the emergence of deep learning techniques, traditional rPPG measurement methods primarily relied on mathematical algorithms to mitigate artifacts and noise from facial video data, thereby obtaining pure rPPG signals. Among these methods, those based on Blind Source Separation (BSS) have proven effective in isolating pulse signals without requiring prior information. A prominent technique within BSS is Independent Component Analysis (ICA), which effectively isolates rPPG signals from three color channels to estimate heart rate [7].

Traditional model-based methods typically utilize the RGB channel information in facial videos to extract rPPG signals. For example, techniques such as Chrominance Mapping (CHROM) [8] and Plane Orthogonal to Skin (POS) [9] successfully extract robust rPPG features from three color channels.

The HR-CNN [10] was the first deep learning-based method to integrate a feature extractor with a heart rate estimator. Compared to traditional approaches, deep learning models have significantly enhanced extraction results and robustness. RhythmFormer [11] employs an end-to-end Transformer approach that explicitly leverages the quasi-periodic nature of rPPG signals to extract them. Liu et al. [12] introduced a Temporal Shift Module (TSM) to capture temporal information, resulting in the development of MTTS-CAN, which addresses the limitations associated with temporal information in 2D convolutional neural networks (CNNs).

In light of the scarcity of large-scale annotated datasets for physiological signals, unsupervised deep learning methods are increasingly gaining traction. Gideon et al. [13] proposed employing contrastive learning for the unsupervised extraction of rPPG signals. Park et al. [14] introduced Fusion ViViT, which utilizes RGB and Near-Infrared (NIR) video frames as inputs and leverages the self-attention mechanism of Transformers to fully capture the spatio-temporal information present in both RGB and NIR videos. Liu et al. [15] proposed an innovative unsupervised approach called rPPG-MAE, which incorporates the Mask Autoencoder technique into rPPG technology.

2.2. Physiological privacy protection

The protection of physiological privacy in facial videos primarily involves three aspects [16,17], i.e., facial identity, facial attributes, and physiological signals. Facial identity is recognized through unique facial features, protecting this information ensures that personal identities are not accessed or identified without authorization. Facial attributes include characteristics such as age, gender, and emotional expressions, safeguarding this information [18] helps prevent misuse or discrimination based on these traits. Protecting physiological signals involves safeguarding sensitive health metrics like heart rate, pulse, blood oxygen levels, and blood pressure, which can be derived from facial analysis [19]. Safeguarding these physiological signals is crucial for maintaining personal privacy and preventing unauthorized health assessments [20].

The application of computer vision techniques for extracting physiological signals has gained significant attention in recent years, bridging the gap between non-invasive health monitoring and ubiquitous computing environments. Physiological signals provide crucial information about an individual's emotions and physical health status; consequently, rPPG technology presents certain privacy leakage risks [16].

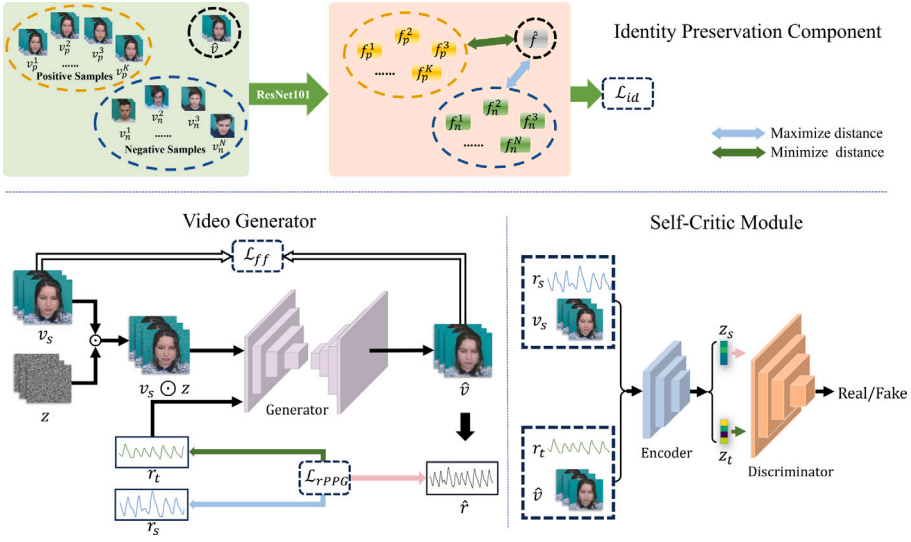


Fig. 1. The overall architecture of the proposed Phys-EdiGAN comprises three main modules, i.e., the Video Generator, the Self-Critic Module, and the Identity Preservation Component, respectively.

However, research focused on the protection of physiological privacy based on rPPG signals remains limited.

To mitigate the disclosure of physiological information in facial videos, Chen et al. [1] employed various motion estimation and elimination methods for video denoising. These methods included techniques for stabilizing camera motion, video de-animation, frequency-domain-based local motion elimination, and Eulerian Motion Magnification (EMM). Their findings indicate that traditional EMM methods are effective for sinusoidal motions but less suitable for non-sinusoidal motions, such as those associated with physiological signals. In contrast, Motion Component Magnification (MCM) algorithms exhibit superior performance in estimating non-sinusoidal motions within videos.

Lin et al. [2] proposed a proactive defense strategy called Signals Hiding (SigH) to mask facial physiological signals. This strategy utilizes BlazeFace for facial region detection, capturing 468 3D facial landmarks that define features such as the tip of the nose and the tops of the cheekbones. Using these landmarks, a ROI is established, excluding the eyes and mouth. For each video frame, the RGB values of the ROI are set to 1, while other areas are set to 0. A sinusoidal signal is then superimposed on this template, which is subsequently blurred using a convolution kernel. Finally, the template sequence is integrated into the source video frames, effectively concealing the physiological signals.

Some studies [21] have shown that adding noise or perturbations can effectively safeguard facial privacy. Chen et al. [3] developed PulseEdit, an algorithm that utilizing negative Pearson Correlation Coefficient constraint to calculate perturbations, so that the perturbation added to the source rPPG signals can be closer to the target rPPG signals. These perturbations are applied as subtle, statistically-driven edits to the skin pixels via random jittering, effectively modifying skin pixels in the video.

In tasks involving the modification of facial physiological signals, it is crucial that only the skin pixels in the facial videos be altered while preserving the background pixels. Sun et al. [4] employed projected gradient descent to ensure compliance with background pixel constraints throughout the iterative process, which includes both gradient descent and projection steps. This methodology guarantees that only skin pixels are modified, thus maintaining the integrity of the background pixels.

Existing methods for protecting physiological privacy signals rely on adding disturbances to the source video. Although this approach effectively interferes with rPPG detection tools, it lacks generalizability and robustness. The deep learning-based model for modifying physiological privacy signals proposed in this paper is applicable to various scenarios and offers faster video editing speeds. Moreover, existing methods do not impose identity constraints on the modified video. The way they add perturbations to the source video may cause distortions in the character's outline or loss of details, leading to the loss of identity information from the source video. The proposed Identity Preservation Component module in this paper effectively addresses this issue, ensuring that the identity information from the source video is well preserved.

3. Proposed method

The architecture of our Phys-EdiGAN consists of three key components: the Video Generator, the Self-Critic Module, and the Identity Preservation Component, as depicted in Fig. 1. The Video Generator receives the source video and target rPPG signals as inputs, with the goal of generating a synthetic video that incorporates the target rPPG signals. The Self-Critic Module takes as input both the source video paired with its corresponding rPPG signals and the generated video paired with its respective rPPG signals. This module learns a critic model that evaluates the intrinsic consistency between the generated facial video and the associated rPPG signals, providing an additional layer of supervision for the learning process. The Identity Preservation Component is designed to retain the identity information of the characters in the video, ensuring the robustness of the network.

3.1. Video generator

Given that both 3D Convolutional Neural Networks (3DCNN) [22] and Transformers [23] effectively capture spatial and temporal dynamics in video data, we utilize them to build the video generator for reconstructing the video and learning the distribution of physiological signals within the video. In particular, 3DCNN is employed to extract spatio-temporal features from the video, which are then concatenated with the features of corresponding rPPG signals. This enables the network to learn the underlying relationships between the video data and the rPPG signals. The Transformer processes these spatial–temporal features to develop higher-level features for generation tasks. Its self-attention mechanism manages information flow over time, capturing long-distance dependencies and resulting in more coherent video frames. Fig. 2 illustrates the architecture of the Video Generator.

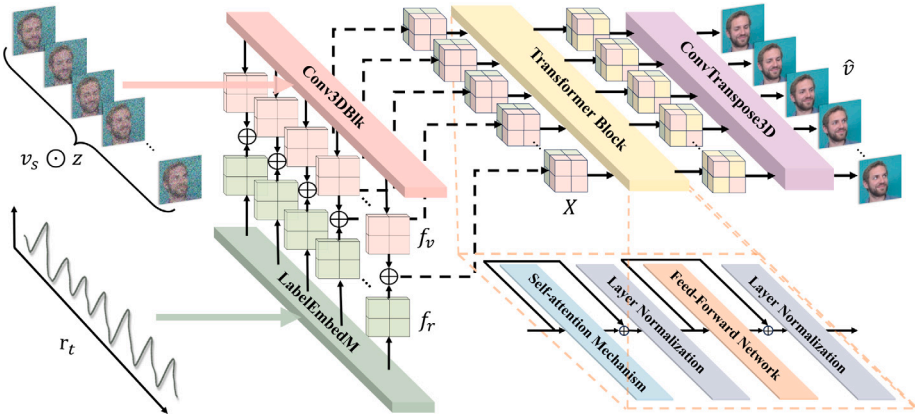


Fig. 2. Architecture of the video generator. It takes as inputs both the weighted noisy video $v_s \odot z$ and the target remote rPPG signals r_t, concatenating their features as inputs to the TransformerBlock to extract spatiotemporal features. These features are subsequently fed into a 3D transposed convolution network, ultimately generating a facial video $\hat{v}$ that contains the target rPPG signals.

Let the source facial video be denoted as $v_s \in \mathbb{R}^{C\times T\times H\times W}$, and the generated facial video as $\hat{v} \in \mathbb{R}^{C\times T\times H\times W}$. The rPPG signals corresponding to the source facial video is represented as $r_s \in \mathbb{R}^T$, while the rPPG signals for the target facial video is denoted as $r_t \in \mathbb{R}^T$. The objective of the video generator is to learn a mapping F: $v_s \to \hat{v}$, such that $\hat{v}$ maintains the same appearance as v_s, but converting the source rPPG signals r_s into the target rPPG signals r_t. As a result, the target video visually looks exactly the same as the source video, but reflects the target rPPG signals. Furthermore, we introduce Gaussian noise $z \in \mathbb{R}^{C\times T\times H\times W}$ along with the input video to eliminate interference in the source rPPG signals, enabling the model to better learn the decoupled representation of rPPG features from other features [24,25]. This also increases the model's generalization ability, allowing it to generate more realistic and natural face videos. Therefore, the video generation process can be denoted as:

$$\hat{v} = G([v_s \odot z, r_t]), \tag{1}$$

where $\odot$ denotes element-wise multiplication, and G represents the video generator, $\hat{v}$ is the generated facial video that contains the target rPPG signal. After being multiplied by noise, the video frames are then used as input to the 3D convolutional blocks (Conv3DBlk), resulting in the generation of the video feature map $f_v \in \mathbb{R}^{C'\times T\times H'\times W'}$, where width W' and height H' are the 1/8 of the source video, and C' is 4 times that of C. Each Conv3DBlk involves a 3D convolutional layer, a 3D Batch Normalization layer and an activation function layer, respectively.

The rPPG signals are represented as a one-dimensional vector of length T, where T denotes the total number of frames in the source facial video. It is encoded through a Label Embedding Module (LabelEmbedM), which consists of a fully connected layer, resulting in an embedding vector $f_r \in \mathbb{R}^{C'\times T\times H'\times W'}$, a same dimensionality of the f_v. Here, the target rPPG signals is treated as a condition to guide the generation process of the video generation network.

Then f_v and f_r are concatenated and treated as the features, which is then fed into the TransformerBlock as below:

$$\begin{cases} f_v = Conv3DBlk(v_s \odot z), \\ f_r = LabelEmbedM(r_t), \\ X = f_v \oplus f_r \end{cases} \tag{2}$$

where $\oplus$ represents vector concatenation, and X is the concatenated features.

When entering TransformerBlock, each concatenated feature first goes through a self-attention mechanism, to generate the matrices of Query (Q), Key (K), and Value (V), respectively. Subsequently, attention scores are calculated by taking the dot product of the queries and keys, followed by scaling these scores by the square root of their dimension.

The self-attention mechanism can be denoted as:

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{256}}\right)V \tag{3}$$

After the self-attention mechanism, the output is added to the source input X through a residual connection, followed by Layer Normalization (LN):

The layer normalization can be denoted as:

$$Y = LN(X + Attention(Q, K, V)) \tag{4}$$

The output Y at each position is then passed through a Feed-Forward Network (FFN), which can be denoted as:

$$FFN(Y) = \max(0, Yw_1 + b_1)w_2 + b_2, \tag{5}$$

where w_1 and b_1 are the weights and bias of the first fully connected layer, while w_2 and b_2 are the weights and bias of the second fully connected layer. FFN further integrates information and extracts core features of the video.

The output of the FFN incorporates a residual connection and LN. To restore the low-dimensional features to the original spatial resolution of the video and generate a video with its original dimensions and temporal sequence, the 3D transposed convolution (ConvTranspose3D) is employed to generate the facial video.

The generated facial video can be obtained through the following formula:

$$\hat{v} = ConvTranspose3D(LN(Y + FFN(Y))) \tag{6}$$

Facial Focus Constraint: Since non-face regions do not contain physiological signals, to eliminate the influence of non-face areas on facial video generation, we incorporate a background mask to enable the model to focus exclusively on generating facial regions that contain physiological signals. To achieve this, facial regions are cropped from the video, and a binary mask is created. In this binary mask, pixel values for the facial regions are set to 1, while all other areas are assigned a pixel value of 0.

We selected the FaceMesh algorithm [26] as our face landmark detector, which can identify 486 3D facial landmarks, allowing for a more detailed delineation of facial regions. Among the 486 key points detected by FaceMesh, we selected 39 facial contour points to outline the facial boundary. In this binary mask M, pixel values within the contour region are set to 1, while pixel values outside this region are set to 0, thus forming a clear binary mask. The facial contours for each frame in the source video are detected to generate the mask M_i. The frames of the source video v_s^i and the target facial video $\hat{v}^i$ are then multiplied with the corresponding mask M_i, resulting in the facial regions of b_s^i and $\hat{b}^i$. The process of the facial focus constraint is shown

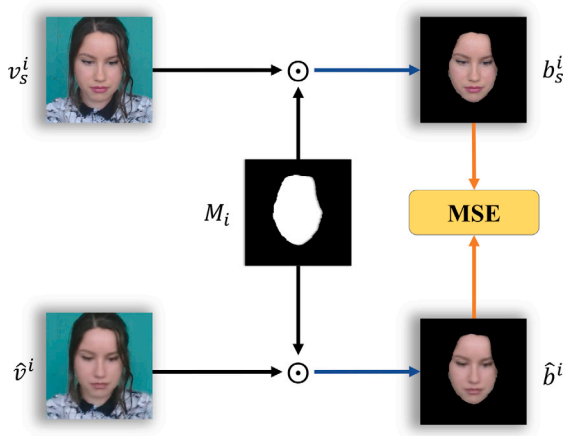


Fig. 3. The process of the facial focus constraint. Given a facial video, the *i-th* frame v_s^i and its generated image $\hat{v}^i$ are both multiplied by the binary mask M_i to obtain the corresponding facial regions b_s^i and $\hat{b}^i$. The MSE loss is used to calculate the facial focus loss between b_s^i and $\hat{b}^i$.

in Fig. 3. Then the Mean Squared Error (MSE) is used to calculate the facial focus loss function $\mathcal{L}_{ff}$:

$$\mathcal{L}_{ff} = \frac{1}{T}\sum_{i=1}^{T}(b_s^i - \hat{b}^i)^2 \tag{7}$$

rPPG Constraint: The objective of our model is to align the rPPG signals derived from generated facial videos with the target rPPG signals, while simultaneously distancing them from the rPPG signals of the source facial video. To achieve this, we introduce an rPPG loss term, $\mathcal{L}_{rPPG}$, which quantifies the divergence between the rPPG signals as:

$$\min \mathcal{L}_{rPPG} = (1 - \rho(\Phi(\hat{v}), r_t)) + (1 + \rho(\Phi(\hat{v}), r_s)), \tag{8}$$

where $\Phi(\hat{v})$ denotes the rPPG signal extracted from the generated facial video $\hat{v}$, with Φ representing the rPPG signal extractor. Φ is implemented using PhysFormer [27], which is based on a Transformer architecture that can globally model the spatiotemporal features in the video, thereby enabling high-precision extraction of physiological signals and providing reliable supervision for the generator. The variable ρ denotes the Pearson Correlation coefficient, which quantifies the similarity between rPPG signals. A value closer to 1 indicates a strong alignment of the rPPG signals from the generated video with the target rPPG signals. In contrast, a value nearer to −1 reflects a significant divergence between the rPPG signals of the generated video and those of the source video.

3.2. Self-critic module

The Self-Critic Module consists of two components, i.e., an Encoder and a Discriminator. The Encoder takes as input both the source facial video v_s paired with its corresponding rPPG signals r_s and the generated facial video $\hat{v}$ paired with the target rPPG signals r_t. It outputs two latent codes z_s and z_t, which are subsequently fed into the Discriminator for the purpose of distinguishing between real and fake. In this process, the model can automatically learn the intrinsic correspondence between the facial video and the rPPG signals, ensuring that the target rPPG signals remains consistent with its corresponding facial video.

Specifically, the source-combination $[v_s, r_s]$ and the target-combination $[\hat{v}, r_t]$ are treated as inputs to the Self-Critic Module. Before feeding these two combinations into the discriminator, the encoder E is employed to extract the latent codes from the inputs, which can be represented as follows: $E([v_s, r_s]) \rightarrow z_s$, $E([\hat{v}, r_t]) \rightarrow z_t$. The encoder E consists of a Conv3DBlk and a LabelEmbedM. The Conv3DBlk extracts video features, while the LabelEmbedM extracts the rPPG signal features. The two feature vectors are then concatenated to obtain the final latent codes.

The role of the generator G is to generate videos $\hat{v}$ that are as realistic as possible and combine them with target rPPG signal r_t, which are then input to the discriminator together with $[v_s, r_s]$ for authenticity evaluation. The discriminator needs to assess the authenticity of these two data combinations and output a probability value indicating the likelihood that the data combination is genuine. When the loss functions of the generator and discriminator stabilize and no longer show significant changes, it can be considered that the model has converged, meaning that training can be stopped.

The training loss of the Self-Critic model can be defined as:

$$\begin{aligned}\mathcal{L}_{sc} =& \mathbb{E}_{x\sim p_{gt}(x)}[\log D(E([v_s, r_s]))] \\ &+ \mathbb{E}_{z\sim p_z(z)}[\log(1 - D(E([\hat{v}, r_t])))]\end{aligned} \tag{9}$$

3.3. Identity preservation component

The training and generation processes of Generative Adversarial Networks (GANs) inherently involve a degree of randomness. During video generation, stochastic elements may result in deviations from anticipated features, consequently altering the identity characteristics of the generated characters. To enhance the consistency of identity in generated facial videos, we have incorporated an Identity Preservation Component (IPC).

In the original dataset, each individual corresponds to a single video. To obtain more positive and negative samples for calculating identity loss, we decompose each video into several short video segments. As a result, each individual corresponds to multiple video segments. During the model training process, assuming the currently generated video is $\hat{v}$, the other video segments with the same identity as video v_p are considered positive samples, while the video segments with different identities from video v_n are considered negative samples. We define the positive samples as $V_p = \{v_p^1, v_p^2, \ldots, v_p^K\}$, and the negative samples as $V_n = \{v_n^1, v_n^2, \ldots, v_n^N\}$. We use ResNet101 [28] to extract features for V_p, V_n and $\hat{v}$, denoted as f_p, f_n and $\hat{f}$ respectively. For each feature vector of positive sample v_p^i, calculate its cosine similarity with $\hat{f}$:

$$s_p^i = \frac{{f_p^i}^T \hat{f}}{\|f_p^i\|\|\hat{f}\|} \tag{10}$$

For each feature vector of negative sample v_n^j, calculate its cosine similarity with $\hat{f}$:

$$s_n^j = \frac{{f_n^j}^T \hat{f}}{\|f_n^j\|\|\hat{f}\|} \tag{11}$$

Then, the identity (ID) loss can be defined as:

$$\begin{aligned}\mathcal{L}_{id} =& \sum_{i=1}^{K}\log(1 + \exp(-\gamma\alpha_p^i(s_p^i - \Delta_p))) \\ &+ \sum_{j=1}^{N}\log(1 + \exp(\gamma\alpha_n^j(s_n^j - \Delta_n))),\end{aligned} \tag{12}$$

where γ is a scale factor, Δ_p and Δ_n represent the margins for positive and negative sample pairs, respectively. The terms α_p^i and α_n^j denote non-negative weighting factors.

3.4. Overall loss

The overall loss is the weighted sum of the facial focus loss $\mathcal{L}_{ff}$, the rPPG loss $\mathcal{L}_{rPPG}$, the self-critic loss $\mathcal{L}_{sc}$, and the ID loss $\mathcal{L}_{id}$. The formal expression is as follows:

$$\min_G \max_D \mathcal{L}_{total} = \lambda_1\mathcal{L}_{rPPG} + \lambda_2\mathcal{L}_{sc} + \lambda_3\mathcal{L}_{id} + \lambda_4\mathcal{L}_{ff}, \tag{13}$$

where λ_1, λ_2, λ_3 and λ_4 are the hyper-parameters to control the contribution of different loss functions to the overall loss. By adjusting these weights, it is possible to balance the importance of each loss during the training process.

4. Experimental results

In this section, we begin by introducing the experimental setup and dataset used in our study. Next, we conduct an ablation study to evaluate and analyze the individual components of Phys-EdiGAN. Following this, we perform both quantitative and qualitative analyses to assess Phys-EdiGAN's capability to modify rPPG signals and the quality of the generated videos.

4.1. Experimental setup and dataset

Implementation: The facial videos are standardized to a fixed resolution of 256 × 256 pixels. The original dataset videos are divided into multiple clips, each lasting 8.53 s (equivalent to 256 frames). During training, we enhance the model's generalization capability and robustness in privacy preservation by randomly selecting an 8.53-s ground-truth rPPG signal from other clip in the dataset as target signal to embed into the currently generated video clip. This data-driven diversity learning strategy ensures broader adaptability and strengthens resistance against adversarial privacy inference attacks. Our experiments were conducted on a system equipped with an Intel i7-13700K CPU, 32 GB of RAM, and an NVIDIA 4090 GPU with 24 GB of memory. The learning rate was set to 0.001, and the training process lasted for 300 epochs. For the hyperparameters, we assigned values of $\lambda_1 = 0.01$, $\lambda_2 = 1$, $\lambda_3 = 0.3$, and $\lambda_4 = 0.7$.

Dataset: We utilized two datasets to train and evaluate our network: UBFC-rPPG [29] and VIPL-HR [30], respectively. The UBFC-rPPG dataset consists of 42 uncompressed 8-bit RGB facial videos recorded at a resolution of 640 × 480 pixels with a frame rate of 30 fps. These videos include diverse identities and varying conditions of sunlight and indoor lighting. Given the relatively small facial areas in these videos, we employed the YuNet [31] face detection model to crop each frame to a 256 × 256 pixel facial region. The VIPL-HR dataset is more extensive, featuring 2378 visible light videos and 752 near-infrared videos of 107 subjects. This dataset encompasses a broad range of facial dynamics, including head movements, diverse lighting conditions (laboratory, dark, and bright), and recordings from various devices (smartphone, webcam, RGB camera). For the two datasets, we divided them into a 70% training set, a 15% validation set, and a 15% test set. To ensure fairness and avoid data leakage, all video clips from the same subject were strictly assigned to the same subset (training, validation and test).

4.2. Ablation study

We performed an ablation study on Phys-EdiGAN to evaluate five variants of the model. (1) Phys-EdiGAN (Base): A standard GAN architecture that generates facial videos directly from the source input video. (2) Phys-EdiGAN (Base + FF): An extension of the base model, enhanced with Facial Focus (FF) loss. (3) Phys-EdiGAN (Base + IPC): The base model augmented with and IPC. (4) Phys-EdiGAN (Base + FF + IPC): A comprehensive model combining both FF loss and the IPC. (5) Phys-EdiGAN (Base + FF + IPC w/o z): A comprehensive model combining both FF loss and the IPC but without noise z.

The quantitative experimental results of the ablation study are presented in Table 1, while the visual experimental results are shown in Fig. 4. By comparing the results of the two configurations, Phys-EdiGAN (Base) and Phys-EdiGAN (Base + IPC), it is evident that incorporating the IPC into the model increases the Identity Similarity (IS) metric by 0.039 and the Peak Signal-to-Noise Ratio (PSNR) by 0.791. The IPC significantly enhances the model's ability to preserve character identity in the generated videos. This component utilizes a pre-trained ResNet101 face recognition model, fine-tuned on a large-scale facial video dataset, to effectively capture distinctive identity features. These features exhibit specific distribution patterns within the feature space, which the IPC leverages to guide the generator in producing images that more closely align with the actual distribution of facial identity features. As illustrated in Fig. 4, the incorporation of the IPC results in video frames that more accurately preserve the shape and contours of the source facial features, thereby substantially improving visual clarity.

By comparing the results of two configurations, Phys-EdiGAN (Base) and Phys-EdiGAN (Base + FF), it is observed that the PSNR and Structural Similarity Index Measure (SSIM) values of Phys-EdiGAN (Base + FF) increased by 4.316 and 0.108, respectively, compared to Phys-EdiGAN (Base). The integration of the Facial Focus loss significantly enhances the quality of the generated videos. Specifically, by applying an MSE loss function to the facial regions, we can quantitatively measure the pixel-wise discrepancies between the generated and reference video frames. This approach guides the generator to minimize these discrepancies, with the squared error metric imposing higher penalties on larger errors. As a result, the generator focuses more intensively on regions exhibiting significant differences, thereby accelerating convergence towards outputs that closely resemble the actual video frames. Additionally, the FF loss ensures that the model concentrates solely on generating the facial region, without altering the background of the source video. Consequently, the video retains more detailed information, leading to substantial improvements in both PSNR and SSIM relative to the base model. As shown in Fig. 4, the generated video's background is indistinguishable from the source, and there is a marked improvement in the color and clarity of the face, with the facial details of the source video largely restored.

A comparison of the results from Phys-EdiGAN (Base + FF + IPC) with those from the other configurations clearly demonstrates that the integration of both FF loss and IPC yields the most significant performance improvements across all evaluation metrics. Specifically, adding both FF loss and IPC to the Phys-EdiGAN (Base) configuration results in a 0.115 increase in SSIM, an 5.559 increase in PSNR, a 0.101 increase in IS, and a 0.096 reduction in LPIPS, compared to Phys-EdiGAN (Base). These improvements highlight the effectiveness of combining both Facial Focus loss and identity preservation techniques in enhancing the overall quality of the generated videos. Fig. 4 provides a visual comparison, where the frames generated by Phys-EdiGAN (Base + FF + IPC) exhibit a high degree of similarity to the source frames, both in terms of facial features and background consistency. The results illustrate that the combination of IPC and FF loss not only enhances the clarity and perceptual quality of the generated frames but also ensures the preservation of both the character's identity and the surrounding environment throughout the video.

To demonstrate the "benefits" of introducing noise z, we compared the Phys-EdiGAN (Base+FF+IPC) with the Phys-EdiGAN (Base+FF+IPC w/o z) variant. The variant without noise had an HR-M-MAE 10.76 higher and an HR-R-MAE 37.67 lower than Phys-EdiGAN (Base+FF +IPC). This suggests that, without the introduction of noise, the model directly fits the strong coupling relationship between the rPPG signal features and other video features, resulting in the model being unable to independently manipulate the rPPG signal during the generation process. Therefore, introducing noise z disrupts the direct correlation between the rPPG features and other video features in the original video, forcing the model to learn a stronger decoupled representation—separating the rPPG signal features from other features (such as identity and expression information). This enhances the model's ability to edit the rPPG signal more effectively.



Fig. 4. The visual results of different ablation variants. The first row of images shows the results generated by the base model. The second and third rows display the results produced by the base model combined with IPC and FF loss, respectively. The fourth row presents the results of the base model incorporating both IPC and FF loss. The fifth row presents the results of the base model incorporating both IPC and FF loss but without noise z. The final row represents the Ground Truth. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 1
Qualitative comparisons of different ablation variants.

Variants	SSIM↑	PSNR↑	LPIPS↓	IS↑	HR-M-MAE↓	HR-R-MAE↑
Phys-EdiGAN(Base)	0.843	30.388	0.122	0.882	24.16	15.93
Phys-EdiGAN(Base + IPC)	0.904	31.179	0.056	0.921	18.49	10.71
Phys-EdiGAN(Base + FF)	0.951	34.704	0.035	0.952	21.24	13.48
Phys-EdiGAN(Base + FF + IPC)	0.958	35.947	0.026	0.983	12.98	43.09
Phys-EdiGAN(Base + FF + IPC w/o z)	0.962	35.980	0.025	0.990	23.74	5.42

Table 2
Results of Phys-EdiGAN on the tasks of rPPG removal and rPPG matching, evaluated separately on the UBFC-rPPG and VIPL-HR datasets.

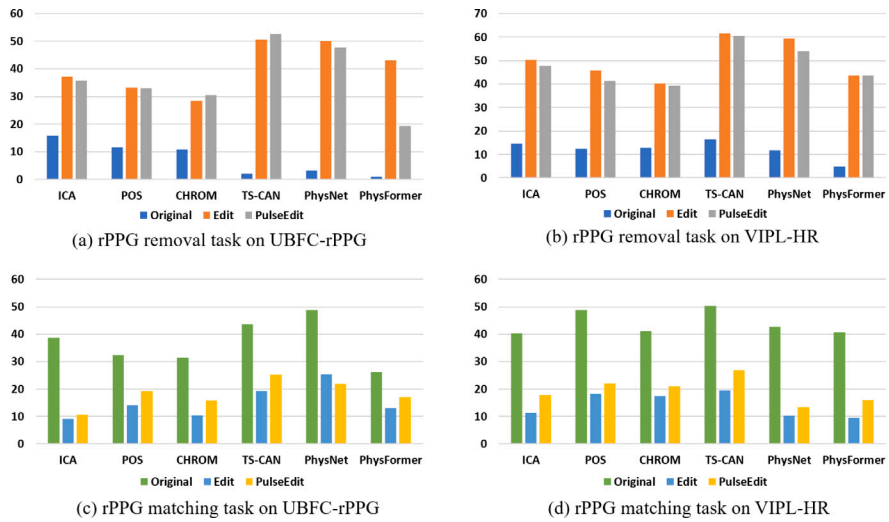
rPPG method	rPPG removal				rPPG matching			
	UBFC-rPPG		VIPL-HR		UBFC-rPPG		VIPL-HR	
	Source HR-B-MAE	Phys-EdiGAN HR-R-MAE	Source HR-B-MAE	Phys-EdiGAN HR-R-MAE	Source HR-B-MAE	Phys-EdiGAN HR-M-MAE	Source HR-B-MAE	Phys-EdiGAN HR-M-MAE
ICA [32]	15.74	37.3	14.52	50.31	38.65	9.14	40.31	11.28
POS [9]	11.64	33.15	12.43	45.73	32.42	14.08	48.93	18.24
CHROM [8]	10.9	28.44	12.76	40.23	31.56	10.51	41.18	17.44
TS-CAN [12]	2.13	50.62	16.52	61.65	43.64	19.26	50.34	19.51
PhysNet [33]	3.16	50.02	11.82	59.32	48.86	25.47	42.71	10.18
PhysFormer [27]	1.06	43.09	4.89	43.61	26.18	12.98	40.63	9.51

4.3. Heart rate modification evaluation

The protection of physiological signal privacy encompasses two distinct sub-tasks: rPPG signal removal and matching. Effective protection is indicated by a substantial difference between the rPPG signals of the source and the generated videos, while strong matching capabilities are shown by minimal discrepancies between the rPPG signals of the generated videos and the target rPPG signals. To quantify the extent to which the source rPPG signals were removed from the facial video, we calculated the Mean Absolute Error (MAE) between the Heart Rates (HR) of the generated and source videos, referred to as HR-Removal-MAE (HR-R-MAE). A higher HR-R-MAE indicates more effective removal of the rPPG signals. We introduce HR-Matching-MAE (HR-M-MAE), which measures the MAE between the HR of the generated video and the target rPPG signals. A lower value indicates better embedding of the target rPPG signals into the source video.

Given the variability in the accuracy and numerical outputs of different rPPG signal extractors, direct comparisons with the Ground Truth (GT) rPPG signals of the source videos may not yield meaningful insights. To evaluate Phys-EdiGAN's ability to modify rPPG signals more fairly, we selected three conventional algorithms—POS [9], CHROM [8], ICA [32] and three deep learning based models—PhysNet [33], TS-CAN [12], and PhysFormer [27] to derive signals from the facial videos, separately. We then calculate a HR-MAE between the signals extracted by each method of the source video and the GT rPPG signals of the source video in each dataset, which is used to address the issue of inconsistent performance among different extraction algorithms and is denoted as HR-Base-MAE (HR-B-MAE).

Due to the limited number of methods available for facial physiological signal editing, and the fact that some methods do not provide source code, making replication challenging, we replicated PulseEdit [1] from the few available methods and compared it with our approach on two datasets (UBFC and VIPL). PulseEdit accurately identifies the facial region and extracts skin signals from multiple subregions. After applying a detrending process, it isolates the perturbation signal and then adjusts the skin pixels within the facial region using randomized dithering, thereby achieving the desired rPPG signal editing.



(a) rPPG removal task on UBFC-rPPG (b) rPPG removal task on VIPL-HR

(c) rPPG matching task on UBFC-rPPG (d) rPPG matching task on VIPL-HR

Fig. 5. Comparison results of Phys-EdiGAN and PulseEdit on the tasks of rPPG removal and rPPG matching across the UBFC-rPPG and VIPL-HR datasets.

The proposed method is evaluated using two key metrics: HR-R-MAE (higher is better) and HR-M-MAE (lower is better). HR-R-MAE measures the MAE between the HR of the generated video and the source video, while HR-M-MAE measures the MAE between the HR of the generated video and the target rPPG signals. Additionally, HR-B-MAE is included as a reference baseline metric for comparison. The results, presented in Table 2, show that regardless of the rPPG signal extraction method used for facial videos, the proposed Phys-EdiGAN achieves consistently high HR-R-MAE values on both the UBFC-rPPG and VIPL-HR datasets. This demonstrates that the rPPG signals from the source video are effectively removed. Furthermore, Phys-EdiGAN achieves consistently low HR-M-MAE values on both datasets, indicating that the target rPPG signals are successfully embedded into the generated facial videos.

Fig. 5 compares the proposed Phys-EdiGAN with PulseEdit on the rPPG removal and rPPG matching tasks using the UBFC-rPPG and VIPL-HR databases. Panels (a) and (b) illustrate the results of the rPPG removal task. While PulseEdit achieves slightly higher HR-R-MAE values than Phys-EdiGAN on the UBFC-rPPG database when extracting rPPG signals from facial videos using CHROM and TS-CAN, Phys-EdiGAN consistently outperforms PulseEdit in all other cases. This phenomenon may stem from the sensitivity of traditional extractors to high-frequency noise, whereas PulseEdit's pixel-level perturbations are better suited for such noise patterns. Panels (c) and (d) present the results of the rPPG matching task. While PulseEdit achieves slightly lower HR-M-MAE values than Phys-EdiGAN on the UBFC-rPPG database when extracting rPPG signals from facial videos using PhysNet, Phys-EdiGAN consistently achieves lower HR-M-MAE values compared to PulseEdit in all of other cases. We speculate that PhysNet has a relatively wide frequency response range. When the target rPPG signal differs significantly from the source signal in the low-frequency band, the generator of Phys-EdiGAN may prioritize optimizing the primary frequency band, temporarily increasing the matching error in the secondary frequency band. This indicates that Phys-EdiGAN demonstrates stronger capabilities in rPPG signal fusion and retargeting compared to PulseEdit.

4.4. Generated video quality evaluation

For video quality evaluation, we used PSNR and SSIM to measure reconstruction errors, and Learned Perceptual Image Patch Similarity (LPIPS) [34] to assess the perceptual quality. Meanwhile, we introduce Spatio-temporal Reduced Reference Entropic Differences (ST-RRED) [35] and Video Multimethod Assessment Fusion(VMAF) to assess the spatio-temporal quality and distortion of the video. Higher PSNR and SSIM values indicate less distortion, while a lower LPIPS value shows greater similarity between the videos, reflecting the robustness of the model. In addition, a lower ST-RRED value represents a smaller degree of distortion in the spatio-temporal domain, and a higher VMAF value represents better video quality.

Current methods for physiological privacy signal protection, such as PulseEdit [3] and Privacy-Phys [4], primarily modify the rPPG signal by adding perturbations to the source video. These approaches only alter pixel values in specific regions of the source video without regenerating it, making comparisons of video quality with our approach unfair. To the best of our knowledge, our method is the first face physiological signal editing approach based on video generation. Therefore, no existing facial physiological signal editing method offers a fair comparison with our approach in terms of video quality.

Facial reenactment, an emerging conditional face generated task, aims to achieve two objectives: (1) transferring the shape of the source face to the target face, and (2) preserving the appearance and identity of the target face. This is similar to our task, which involves transmitting the rPPG signal to the target face while maintaining the target face's appearance and identity. In tasks that require high-fidelity facial reconstruction from monocular videos, models such as HFA-GP [36], DFRF [37], and AD-NeRF [38] have demonstrated the ability to reconstruct facial videos and generate high-quality talking head videos from limited facial images of an individual.

To evaluate the performance of Phys-EdiGAN in video generation, we compared the videos generated by Phys-EdiGAN with those generated by the three methods mentioned above. All methods were trained and tested using the data from the HFA-GP experiment. The quantitative comparison results of the four methods are shown in Table 3, where it is evident that Phys-EdiGAN outperforms the other three methods in terms of SSIM, PSNR, LPIPS, ST-RRED and VMAF values, indicating that Phys-EdiGAN performs well in spatio-temporal domain features and video quality preservation in video reconstruction. Additionally, the visual comparison results of the four methods are shown in Fig. 6. It is clear that, in terms of detail generation for video synthesis, the video processed by Phys-EdiGAN exhibits superior detail compared to the competing methods. Specifically, we compared regions with rich local textures, such as the lip shapes and teeth. Compared to methods like HFA-GP and DFRF, our method generates clearer teeth and lip contours that are closer to the source image, demonstrating that Phys-EdiGAN performs better in preserving details.

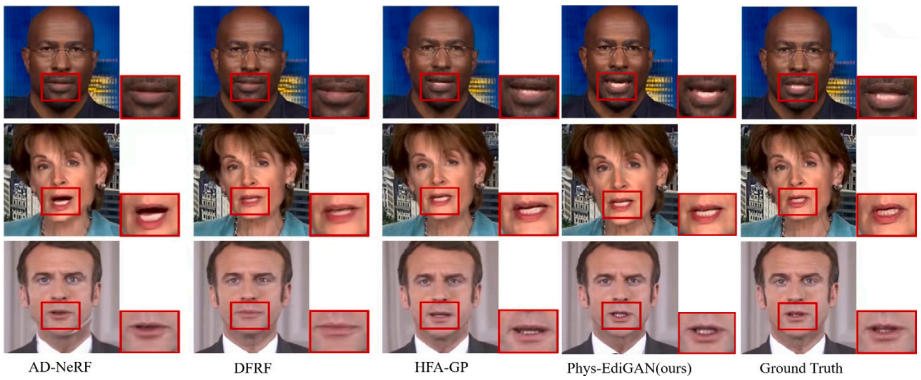


Fig. 6. The comparison results of Phys-EdiGAN with other popular methods for the task of face reenactment. The mouth and teeth regions are enlarged to showcase more details.

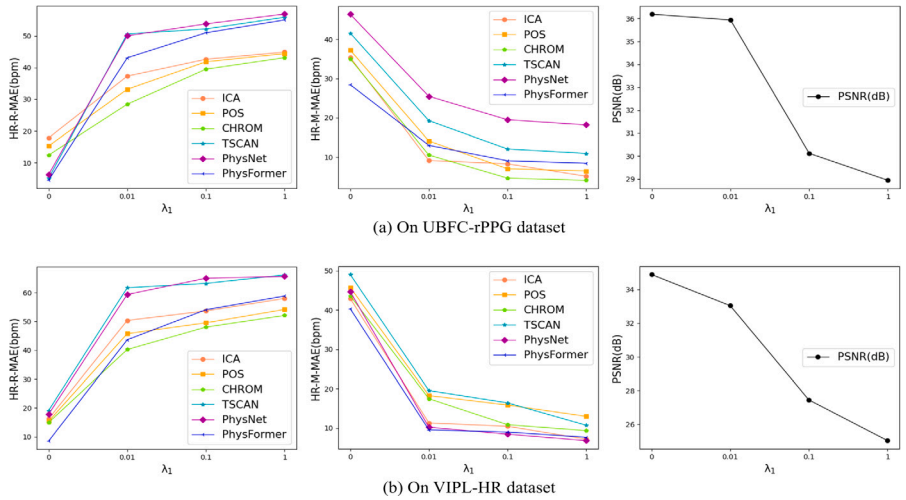


Fig. 7. The line graph for the selection of parameter λ_1. It shows the trend of rPPG editing effect and video generation quality as the parameter λ_1 changes.

Table 3
Quantitative comparison results of video quality generated by different methods.

Method	SSIM↑	PSNR↑	LPIPS↓	ST-RRED↓	VMAF↑
AD-NeRF [38]	0.934	29.69	0.057	0.0819	49.55
DFRF [37]	0.939	30.23	0.042	0.0625	67.83
HFA-GP [36]	0.957	32.57	0.035	0.0381	82.17
Phys-EdiGAN (ours)	**0.958**	**35.94**	**0.026**	**0.0270**	**91.88**

4.5. Supplementary experiment evaluation

The selection of parameter λ_1: We found that the video quality decreased after introducing $\mathcal{L}_{rPPG}$. Therefore, we first iteratively adjusted λ_2-λ_4 using pre-training experiments on the validation dataset to optimize the trade-offs between SSIM, PSNR, LPIPS, and IS. The optimal selection of these hyperparameters ensures the highest video quality before adjusting λ_1. After fixing λ_2-λ_4, we focused on selecting λ_1 to achieve the best balance between rPPG editing and video quality. In our experiments, we tested λ_1 values of {0, 0.01, 0.1, 1} to evaluate their effect on model performance.

To determine the optimal λ_1, we computed the PSNR, HR-R-MAE, and HR-M-MAE values for videos generated under different λ_1 values (refer to Fig. 7). As λ_1 increased, the PSNR of the videos gradually decreased, indicating a loss in video quality, particularly for λ_1 values above 0.01 where the rate of decline in PSNR became more pronounced. When calculating the HR-R-MAE and HR-M-MAE values, we employed 6 rPPG extraction methods: POS, CHROM, ICA, PhysNet, TS-CAN and PhysFormer. We found that although the values of the two metrics differ when computed using different methods, their trend patterns remain consistent. As λ_1 increases, HR-R-MAE increases, indicating a greater difference between the rPPG signals of the source and generated videos, suggesting improved signal removal. However, the improvement slows down when λ_1 exceeds 0.01. Conversely, HR-M-MAE decreases with increasing λ_1, reflecting a reduced difference between the source and generated rPPG signals, with the enhancement slowing down once λ_1 exceeds 0.01. Balancing the objectives of effective rPPG signal removal, accurate modification, and maintaining video quality, we selected a λ_1 value of 0.01 for subsequent experiments.

Running speed: Comparative performance evaluations show that PulseEdit achieves a processing speed of 13 fps with GPU acceleration. In contrast, Privacy-Phys, which uses a 3D CNN architecture optimized for GPU, operates at 42 fps. However, its throughput is limited due to reliance on iterative projected gradient descent calculations. In comparison, our method, fully optimized for GPU acceleration, significantly outperforms these models, achieving a video processing speed of 304 fps with a parameter count of 3.61M. This demonstrates a considerable advantage in processing efficiency, establishing a new benchmark for speed in video processing tasks related to physiological privacy protection.

The distribution of rPPG signal: We validated the effectiveness of Phys-EdiGAN in modifying rPPG signals by quantifying the pixel differences between frames of the source and generated videos. Fig. 8 presents the pixel difference heatmap of the R, G and B channels between the video frames generated by Phys-EdiGAN and the real video frames, where red indicates a large difference and blue indicates a small difference. To avoid the randomness of the results, we fused all the frame heatmaps of each person to obtain the final heatmap.

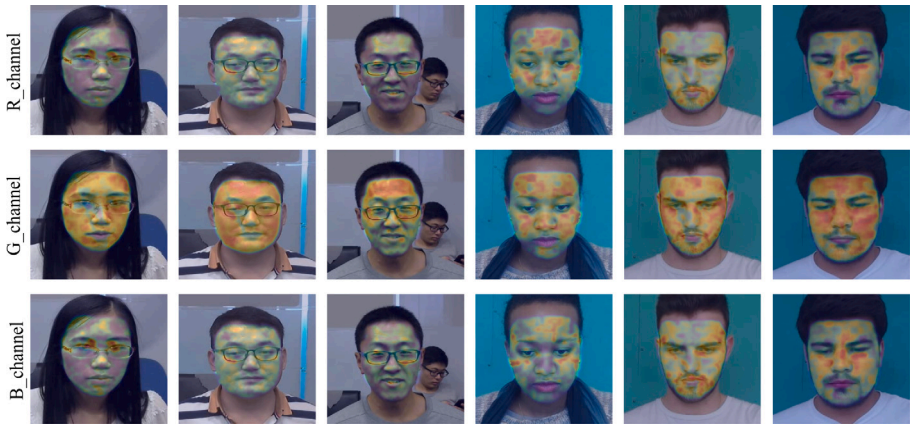


Fig. 8. The pixel difference heatmap of the R, G, and B channels between the video frames generated by Phys-EdiGAN and the real video frames. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Research by Wang et al. [9] indicates that rPPG signals primarily reside in the green (G) channel of facial images. As shown in Fig. 8, the most significant pixel changes occur in this channel, confirming the effectiveness of our method in manipulating the color channel distribution of PPG signals. Additionally, Fig. 8 clearly demonstrates that the rPPG signals affected by Phys-EdiGAN are primarily distributed in the forehead and cheek regions of the face [39], which is consistent with the common knowledge of physiological signal distribution. This observation validates the effectiveness and rationale of Phys-EdiGAN in modifying facial physiological signals.

5. Conclusion

In this paper, we introduce Phys-EdiGAN, a novel and effective method for editing physiological signals in facial videos, leveraging the power of Conditional Generative Adversarial Networks (CGANs). It enables precise, customizable modifications of rPPG signals while preserving facial identity and video quality, addressing the shortcomings of prior methods such as pixel-level noise and manual adjustments. Through extensive experiments on benchmark datasets, we have shown that Phys-EdiGAN can successfully remove, alter, or retarget rPPG signals without affecting video integrity. The integration of an Identity Preservation Component further ensures applicability in privacy-sensitive scenarios. Although the current model performs well on tasks involving rPPG signal removal, modification, and retargeting in facial videos, future work will focus on reducing its computational cost, enhancing generalization across environments, and expanding to other biosignals like respiratory rate and blood pressure to strengthen privacy protection.

CRediT authorship contribution statement

Xiaoguang Tu: Writing – original draft, Formal analysis, Conceptualization. **Zhiyi Niu:** Methodology. **Juhang Yin:** Methodology, Investigation. **Yanyan Zhang:** Validation, Resources. **Ming Yang:** Formal analysis, Conceptualization. **Lin Wei:** Project administration, Conceptualization. **Yu Wang:** Visualization, Resources. **Zhaoxin Fan:** Visualization, Validation. **Jian Zhao:** Visualization, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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